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ORIGINAL RESEARCH

Improved Postexercise Rehydration With a Milk Permeate-Based Sports Drink

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This study evaluated the rehydration efficacy of a commercially available milk permeate-containing sports drink with high osmolality (MPSD) compared with a traditional sports drink (TSD), a high-potassium sports drink (HKSD), and water (W). Twelve young, healthy adults (four women) underwent a 2-hr exercise-induced dehydration protocol in the heat (29.9 °C ± 1.5 °C) until losing 2% of body weight, followed by a 4-hr rehydration period. Participants consumed on four separate occasions one of the following beverages: (a) TSD (Na^+: 20 mmol/L, K^+: 3.6 mmol/L, osmolality: 385 mmol/kg); (b) HKSD (Na^+: 3.6 mmol/L, K^+: 37.9 mmol/L, osmolality: 364 mmol/kg); (c) MPSD (Na^+: 21 mmol/L, K^+: 28.7 mmol/L, osmolality: 576 mmol/kg); or (d) plain water (W). Fluids (150% of body weight loss) were administered in four doses every 15 min during the first hour. Whole-body net fluid balance remained negative, but MPSD resulted in significantly higher net fluid balance at 180 min (–0.38% ± 0.3%) and 240 min (–0.66% ± 0.3%) than in other trials ($p < .05$). MPSD also led to lower cumulative urine volume (1,268 ± 173 ml) than HKSD (1,796 ± 459 ml), TSD (1,493 ± 211 ml), and W (1,565 ± 501 ml; $p < .05$). Osmotic excretion in MPSD was significantly higher than in all trials at 120, 180, and 240 min ($p < .05$), and free water clearance was lower at 180 and 240 min ($p < .05$). TSD promoted greater plasma volume restoration (1.5% ± 5%) than MPSD (–3.7% ± 6.5%) and HKSD (–5.5% ± 7.8%) at 60 min ($p < .05$). A milk permeate-based sports drink is more effective for postexercise rehydration than TSD, HKSD, and water when consumed at 150% of the body weight lost.

***Keywords**:* hydration, fluid retention, electrolyte

Exercise in a hot environment increases body temperatures, facilitating heat dissipation primarily through sweat evaporation (Galloway & Maughan, 1997). Without sufficient fluid replacement, this results in measurable body water loss (dehydration), impairing thermoregulation (Casa et al., 2005). Fluid intake during exercise is often insufficient, and most athletes experience postexercise dehydration (Arnaoutis et al., 2015). As a result, there is a need for efficient postexercise rehydration.

Rehydration depends on gastric emptying, intestinal absorption via solute transport, and fluid retention across body compartments, with the critical factors including beverage volume, electrolytes (particularly sodium), carbohydrates, energy content, and osmolality (Evans et al., 2017). Shirreffs et al. (1996) demonstrated that ingesting 150% of body weight lost through sweat is sufficient to restore fluid balance, provided that sodium intake adequately replaces sodium lost in sweat (Shirreffs et al., 1996). Given the wide variability in sweat sodium loss (41.7 ± 27.9 mmol/hr) and concentration (30.5–67.7 mmol/L) among individuals (Barnes et al., 2019), no single beverage formulation can universally meet hydration needs under all conditions.

Sports drinks are widely used as rehydration tools and are typically isotonic, containing carbohydrates, electrolytes, and fluids. Combining carbohydrates and sodium enhances fluid absorption in the small intestine via sodium–glucose cotransport and promotes fluid retention in body compartments (Kamijo et al., 2012). However, sodium concentration in most sports drinks is at the lower end of the typical sweat sodium range (20–30 mmol/L), which may be insufficient for effective rehydration, particularly beyond the first 2 hr (Maughan & Leiper, 1995; Merson et al., 2008; Osterberg et al., 2010; Shirreffs, Aragon-Vargas, et al., 2007; Shirreffs & Maughan, 1998). Studies suggest that at least 40 mmol/L of sodium in a beverage is required to observe a significant difference in net fluid balance (NFB) restoration compared with water (Maughan & Leiper, 1995; Merson et al., 2008). A hypertonic solution (667–691 mmol/L) with a carbohydrate concentration of 10%–12% may help offset lower sodium content by slowing gastric emptying and intestinal absorption (Clayton et al., 2014; Evans et al., 2009; Osterberg et al., 2010); however, this concentration is considerably higher than that typically found in sports drinks. In addition, adding potassium, a major intracellular cation, to increase water retention has not shown added benefits over water or carbohydrate electrolyte beverages (James & Shirreffs, 2015; Pérez-Idárraga & Aragón-Vargas, 2014; Shirreffs, Aragon-Vargas, et al., 2007).

Milk has been investigated as an alternative sports drink for rehydration due to its high electrolyte content, proteins, minerals, and comparable carbohydrate concentration. Maughan et al. (2016) reported that both skim and whole milk demonstrated a beverage hydration index comparable with that of a conventional sport drink under euhydrated conditions (Maughan et al., 2016). Furthermore, skim and whole milk were shown to be more effective than the traditional sport drink (TSD) in restoring fluid balance following exercise-induced dehydration (Desbrow et al., 2014; Shirreffs, Watson, et al., 2007), or at least showed comparable effectiveness (Aragón-Vargas et al., 2024; Seery & Jakeman, 2016; Watson

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et al., 2008), despite concerns about palatability and gastrointestinal comfort. Compositional differences between milk and conventional sports drinks make it challenging to identify the specific component responsible for milk's superior hydration. Existing studies suggest that milk's high energy density (Aragón-Vargas et al., 2024; Desbrow et al., 2014; Maughan et al., 2016) and/or the presence of milk protein (James et al., 2011, 2013) may slow gastric emptying and delay fluid absorption into circulation, thereby preserving plasma osmolality, reducing diuresis, and enhancing fluid retention.

To improve palatability and digestibility while reducing energy density and preserving hydration benefits, researchers have developed milk permeate, a byproduct of milk ultrafiltration (Beucler et al., 2005). The milk permeate solution (MPS) is lactose free, contains glucose and galactose, lacks fat and protein, and has a mineral profile similar to milk, with sodium levels comparable with those of TSDs, elevated potassium content, and a higher osmolality (Berry et al., 2022). Under thermoneutral conditions, a sports drink formulated with MPS showed a higher beverage hydration index compared with a conventional sports drink (Berry et al., 2020). The researchers speculated that the elevated beverage hydration index of MPS may be due to its higher content of osmotically active substances, which could enhance fluid retention by delaying gastric emptying and/or intestinal absorption (Berry et al., 2020).

The rehydration efficacy of MPS as a sports drink under environmentally stressful conditions, such as exercising in the heat, remains unclear. To date, only one study has investigated the effects of partial rehydration with an MPS-based sport drink (replacing 75% of body weight loss) on subsequent cycling time-trial performance in the heat, reporting no differences in NFB among MPS, TSD, and water (Berry et al., 2021). Therefore, comparing the hydration efficacy of this hypertonic novel beverage with that of isotonic conventional sports drinks containing different electrolyte compositions and water when consuming sufficient fluid volumes over a full rehydration period is warranted. If the MPS-based sport drink results in greater NFB, it may serve as an effective rehydration option with a lower caloric content and rich in minerals similar to those found in milk.

The primary objective of this study was to evaluate the efficacy of a commercially available MPS-based sports drink compared with commercial sports drinks with different sodium or potassium concentrations and water by measuring fluid and electrolyte balance. It was hypothesized that the MPS-based sports drink would have a greater NFB than other beverages. The milk permeate beverage was also expected to sustain a higher plasma osmolality during rehydration.

Methods

Experimental Design

In this randomized crossover design study, 12 participants completed four identical trials. To observe the effects of different commercial beverages on rehydration, subjects underwent a 2-hr exercise-induced dehydration to 2% of their body weight, followed by a 4-hr rehydration. Block randomization was used to determine each participant's trial/drink order. Participants remained blinded to the drinks used in the study until they completed all four trials.

Participants

A total of 12 healthy young women and men completed this crossover, randomized controlled trial. The characteristics of the volunteers in this study appear in Table 1. Eligibility criteria for participation included training at least twice per week, being 18–55 years old, having a stable weight for >2 months (±2.5 kg fluctuation), and having a body weight of at least 50 kg. Exclusion criteria included night shift work, chronic diseases, medication that could affect water metabolism, and inability to participate in the study. Participants were recruited through word of mouth, flyers, and online advertisements.

Table 1 Participants Characteristics

	Male	Female	Total
Sample size, *n*	8	4	12
Age, years	23.6 ± 1.3	22.5 ± 0.7	22.8 ± 1.9
Height, m	1.78 ± 0.06	1.68 ± 0.01	1.75 ± 0.08
Weight, kg	77.6 ± 10.2	66.1 ± 2.5	73.7 ± 11.3
Body mass index, kg/m^2	24.6 ± 3.1	23.6 ± 3.1	24 ± 3.3
Body fat, %	22.1 ± 5.2	30.1 ± 13.4	24.9 ± 9.1
Fat-free mass, kg	60.5 ± 8.0	45.0 ± 5.4	55.4 ± 10.4
Total body water, L	44.3 ± 5.9	31.0 ± 5.6	40.5 ± 7.6
Extracellular fluid, L	18.7 ± 2.3	13.9 ± 1.6	17.1 ± 3.1
Intracellular fluid, L	25.6 ± 4.1	17.4 ± 3.6	23.5 ± 4.8

Note. Values are represented as mean ± *SD*.

An a priori power analysis was conducted using G*Power based on the difference in NFB between a carbohydrate–electrolyte solution and water reported by Shirreffs et al. (2007) (Shirreffs, Watson, et al., 2007). A sample of 11 participants was required to detect a significant difference between milk permeate sports drink (MPSD) and water with 80% power and an alpha of .05. To account for potential attrition, one additional participant was recruited.

The study was approved by the Arizona State University review board for human experimentation in accordance with the Helsinki Declaration of 1975, as revised in 1983. This trial was registered before the onset of subject recruitment at http://www.clinicialtrials.gov/ as NCT06415409.

Study Controls

Experiments were separated by at least 4 days. Female subjects were tested during the early follicular phase of their menstrual cycle to control for the effect of reproductive hormones on fluid balance. Participants were instructed to avoid exercise training the day prior to each trial. They completed a food and fluid diary for the 24 hr preceding the first trial and were asked to replicate the same intake before each subsequent trial. Subjects consumed a provided meal the evening before each trial and were instructed to consume an adequate amount of fluid to ensure optimal hydration on the morning of the experimental day. After dinner, subjects consumed only plain water for 10 hr before each trial. On the morning of each visit, participants were instructed to consume 500 ml of water, finishing no later than 30 min prior to arrival at the laboratory. These standardization procedures were implemented to minimize variability in pretrial hydration and nutritional status.

Study Protocol

Participants began each trial after a minimum of 10 hr of fasting. Upon arrival, subjects provided an initial urine sample to ensure adequate hydration (urine specific gravity ≤ 1.020). Following this, a phlebotomist inserted an intravenous catheter into an antecubital vein, and breakfast was provided. Subjects consumed a plain bagel, 30 ml of cream cheese, and 300 ml of orange juice.

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After 20 min of rest, a baseline blood sample was collected at room temperature, the participants emptied their bladders, and a baseline euhydrated body weight was recorded.

Dehydration

After the baseline blood sample, the dehydration process started with mild exercise at 70% of their age-predicted maximum heart rate and heat exposure (29.9 °C ± 1.5 °C), targeting a body weight loss of 2%. Participants completed four bouts of 25-min exercise with a 5-min rest between each bout, alternating between a cycling ergometer and walking on a treadmill until a 2% body weight loss or 2 hr of exercise was achieved.

Rehydration

After dehydration, subjects rested for 20 min in a thermally comfortable environment (21 °C), and a baseline, dehydrated blood sample was collected before starting the 4-hr rehydration period. During the first hour of rehydration, subjects ingested 150% of their body weight loss in four equal doses every 15 min (Shirreffs & Maughan, 1998). For the remaining 3 hr, subjects remained seated in a comfortable chair. Participants consumed one of four drinks; two conventional sports drinks (Gatorade and BodyArmor), a milk permeate-based sports drink (GoodSport), and bottled water (AquaFina) as the control. Gatorade was noted for its standard electrolytes and comparable sodium content (TSD, 6% glucose/sucrose), BodyArmor for its higher potassium content (HKSD, 6.1% glucose/sucrose), and GoodSport for its comparable sodium and potassium levels with higher osmolality (MPSD, 4% glucose/galactose and erythritol). Table 2 presents the beverage composition.

Biochemical Analysis

Eight blood samples were collected at the euhydrated and dehydrated baselines and at 30, 60, 90, 120, 180, and 240 min of the rehydration period without stasis via a venous catheter. All blood samples were collected in a seated position at the same room temperature and without stasis to avoid changes in plasma volume that could affect the results. Before the blood draw, subjects remained in the same position for at least 20 min to ensure bodily fluid stabilization. Blood samples were taken prior to subjects using the bathroom. All samples were analyzed for osmolality, total plasma protein, glucose, sodium, potassium, and chloride. Plasma volume changes were calculated based on hematocrit and hemoglobin changes with the Dill–Costill equation (Dill & Costill, 1974). Plasma osmolality was measured using a fully automated and multisample osmometer (A2O Advanced Automated Osmometer, Advanced Instruments Inc.), and plasma glucose, lactate, sodium, potassium, and chloride were analyzed using an ABL90 Flex. Urine aliquots were analyzed for osmolality (A2O Advanced Automated Osmometer, Advanced Instruments Inc.), specific gravity (PAL-10S, Atago), sodium, potassium, and chloride (Beckman Coulter AU 480 Clinical Chemistry Analyzer).

Body Fluid Compartments

Body fluid compartments for intra- and extracellular fluids were assessed using a multifrequency bioelectrical impedance spectroscopy unit (Impedimed SFB7). The electrodes were placed on the participant's nondominant posterior hand and superior foot.

Perceptual Data

Perceptual data for thirst, stomach fullness, and taste were assessed using a visual analog scale. These data were collected immediately before each blood sample, with each perception presented individually. The visual analog scale consisted of a 175-mm line with an anchor at 0 mm ("not at all") and a second anchor on the 125-mm mark labeled "extremely," defined as "the most they had ever felt in their life" (Rolls et al., 1980).

Statistical Analysis

All statistical analyses were conducted using JMP 18 (SAS Institute Inc.). Descriptive data are presented as mean ± *SD*. A repeated-measures analysis of variance was performed to examine differences in NFB, urine volume, osmolality, osmotic excretion, plasma volume, plasma osmolality, and plasma electrolytes across four conditions (MPSD, TSD, HKSD, and W) and multiple time points. When a significant main effect or Beverage × Time interaction was detected, post hoc analyses were conducted using Tukey's HSD correction to adjust for multiple comparisons. Missing data were imputed using linear regressions to estimate the most probable values. Statistical significance was set at $p < .05$.

Results

Net Fluid Balance

Figure 1 illustrates the changes in body weight over time across the different trials. Dehydration (~2% of body weight) did not differ between trials ($p > .05$). A significant main effect of beverage, $F(3, 264) = 5.9$, $p = .0006$; time, $F(5, 264) = 321.6$, $p < .001$; and Beverage × Time interaction, $F(15, 264) = 2.2$, $p = .0073$, was observed for NFB. MPSD preserved significantly more body weight than TSD, HKSD, and W at 180 and 240 min ($p < .05$). TSD also showed a greater NFB than HKSD at 240 min ($p < .05$).

Table 2 Experimental Drinks Composition

Drink	Sodium, mmol/L	Potassium, mmol/L	Carbohydrates, g/L	Magnesium, mmol/L	Energy, kJ/L	Osmolality, mmol/kg H_2O
HKSD	3	37.9	61.3	6.09	1,063	364
TSD	20	3.6	60	0	993	385
MPSD	21	28.7	44	2.46	711	576
W	0	0	0	0	0	0

Note. MPSD contains chloride (27.63 mmol/L), magnesium (2.47 mmol/L), iodine and selenium (<1.0 mmol/L), and erythritol (160 mmol/L). HKSD = high-potassium sports drink, TSD = traditional sports drink, MPSD = milk permeate sports drink, W = water.

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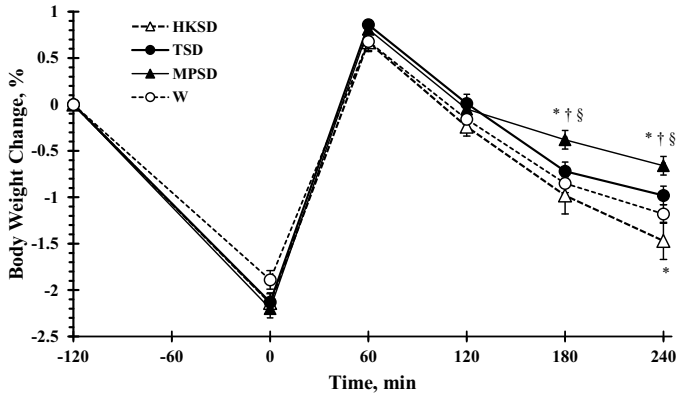

Figure 1 — Body weight change across time following rehydration. Values represent mean ± *SE*. Measurements were obtained at baseline (−120 min), immediately following dehydration (0 min), and at intervals during the subsequent 240-min rehydration period. *Statistically significant difference from TSD. †Statistically significant difference from HKSD. §Statistically significant difference from W. TSD = traditional sports drink; HKSD = high-potassium sports drink; W = water.

Urine Output

A significant main effect of beverage, $F(3, 263) = 3.01$, $p = .0309$; time, $F(5, 263) = 42.6$, $p < .001$; and Beverage × Time interaction, $F(15, 263) = 1.8$, $p = .0308$, was observed for urine volume. A peak in urine volume occurred at 120 min in all groups, followed by a steep decline in MPSD at 180 min, significantly lower than all other beverages ($p < .05$; Figure 2A). HKSD resulted in the highest urine volume at 240 min, significantly higher than MPSD and TSD ($p < .05$). Cumulative urine volume increased over time across all beverages (Figure 2B; $p < .001$). A main effect of beverage was detected, $F(3, 176) = 6.6$, $p = .0003$. MPSD led to lower cumulative urine output compared with HKSD and W, and HKSD exceeded both TSD and W ($p < .05$).

Urine Osmolality and Urine Osmotic Excretion

All fixed effects reached significance for urine osmolality, beverage: $F(3, 263) = 15.3$, $p < .001$; time: $F(5, 263) = 15.9$, $p < .001$; Beverage × Time: $F(15, 263) = 4.9$, $p < .001$. MPSD elicited greater osmolality than W at 120 min and higher values than HKSD, TSD, and W at 180 and 240 min ($p < .05$; Table 3). Osmotic excretion was influenced by beverage, $F(3, 263) = 9.7$, $p < .001$; time, $F(5, 263) = 3.1$, $p = .0102$; and Beverage × Time, $F(15, 263) = 4.8$, $p < .001$. MPSD produced significantly higher values than all other beverages at 120, 180, and 240 min (Figure 3A and 3B; $p < .05$). Significant effects were observed for cumulative osmotic excretion by beverage, $F(3, 175) = 63.5$, $p < .001$; time, $F(3, 175) = 119.2$, $p < .001$; and Beverage × Time, $F(9, 175) = 7.9$, $p < .001$. HKSD resulted in greater cumulative osmotic excretion than TSD and W at 180 and 240 min (Figure 3B; $p < .05$).

Free Water Clearance and Osmotic Clearance

Free water clearance was significantly affected by beverage, $F(3, 151) = 21.5$, $p < .001$; time, $F(3, 151) = 45.8$, $p < .001$; and Beverage × Time interaction, $F(9, 151) = 4.3$, $p < .001$. MPSD showed a consistent decline during rehydration, with significantly lower values than all other beverages at 180 and 240 min (Figure 4A; $p < .05$). Osmotic clearance was significantly affected by beverage,

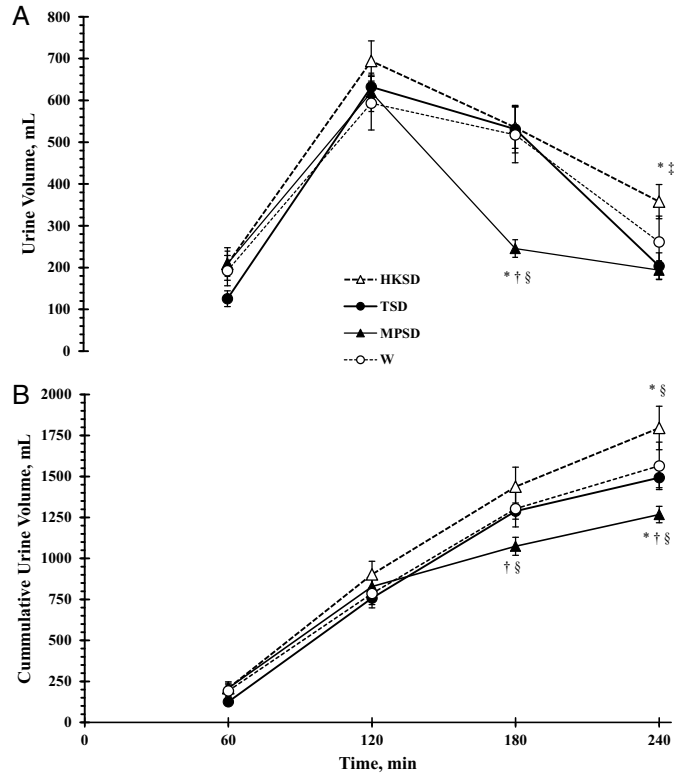

Figure 2 — Urine volume (A) and cumulative urine volume (B) from immediately after dehydration through the entire 240-min rehydration period. Values represent mean ± *SD*. *Statistically significant difference from TSD. †Statistically significant difference from HKSD. §Statistically significant difference from W. ‡Statistically significant difference from MPSD. TSD = traditional sports drink; HKSD = high-potassium sports drink; W = water; MPSD = milk permeate sports drink.

$F(3, 151) = 23.1$, $p < .001$. MPSD yielded higher values than all other beverages (Figure 4B; $p < .05$).

Urine Electrolytes

Urinary Na^+ concentration was significantly influenced by beverage, $F(3, 260) = 8.4$, $p < .001$; time, $F(5, 260) = 10.9$, $p < .001$; and Beverage × Time interaction, $F(15, 260) = 3.6$, $p < .001$. MPSD yielded higher Na^+ levels than HKSD, TSD, and W at 180 and 240 min (Table 3; $p < .05$). HKSD was also elevated relative to TSD at 180 min ($p < .05$). There were main effects of beverage, $F(3, 165) = 44.1$, $p < .001$; time, $F(3, 165) = 3.6$, $p = .014$; and Beverage × Time, $F(9, 165) = 2.5$, $p = .009$ interaction for Na^+ excretion. TSD resulted in lower sodium excretion than MPSD and HKSD, and W resulted in lower Na^+ excretion than MPSD (Table 3; $p < .05$). Similarly, cumulative Na^+ excretion (Table 3) showed a main effect of beverage, $F(3, 165) = 63.7$, $p < .001$; time, $F(3, 165) = 128.9$, $p < .001$; and Beverage × Time interaction, $F(9, 165) = 8.1$, $p < .001$.

Urinary Cl^- concentration followed a similar pattern, with significant effects of beverage, $F(3, 259) = 14.7$, $p < .001$; time, $F(5, 259) = 15.3$, $p < .001$; and Beverage × Time interaction, $F(15, 259) = 5.5$, $p < .001$. MPSD showed higher Cl^- levels than all other beverages at 180 and 240 min (Table 3; $p < .05$). There was a main effect of beverage, $F(3, 165) = 75.7$, $p < .001$; time, $F(3, 165) = 4.5$, $p = .004$; and a Beverage × Time interaction, $F(9, 165) = 5.1$, $p < .001$ for Cl^- excretion (Table 3). MPSD yielded greater Cl^- excretion than other beverages ($p < .05$), followed by HKSD

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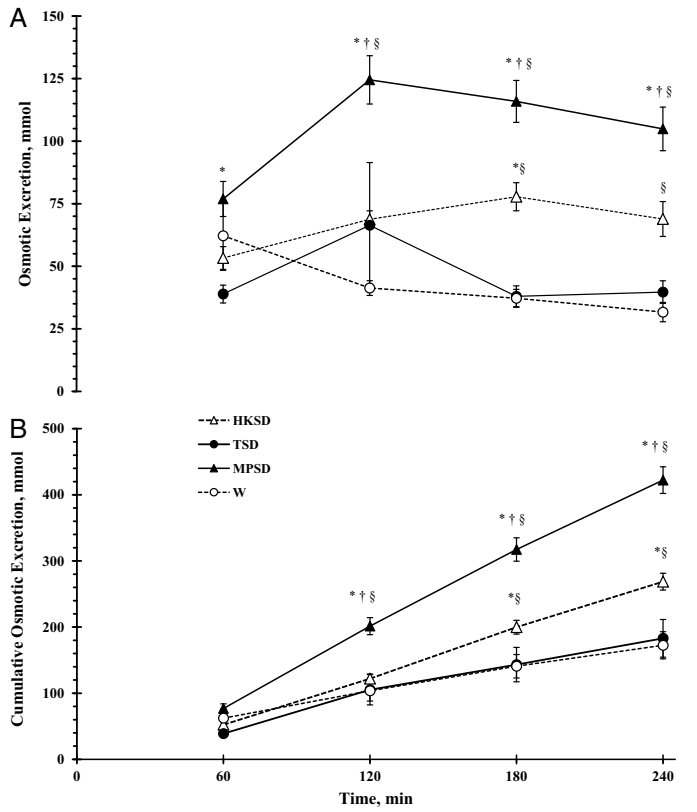

Figure 3 — Urine osmotic excretion (A) and cumulative osmotic excretion (B) during the 240 min of rehydration. Values represent mean ± *SD*. *Statistically significant difference from TSD. †Statistically significant difference from HKSD. §Statistically significant difference from W. TSD = traditional sports drink; HKSD = high-potassium sports drink; W = water.

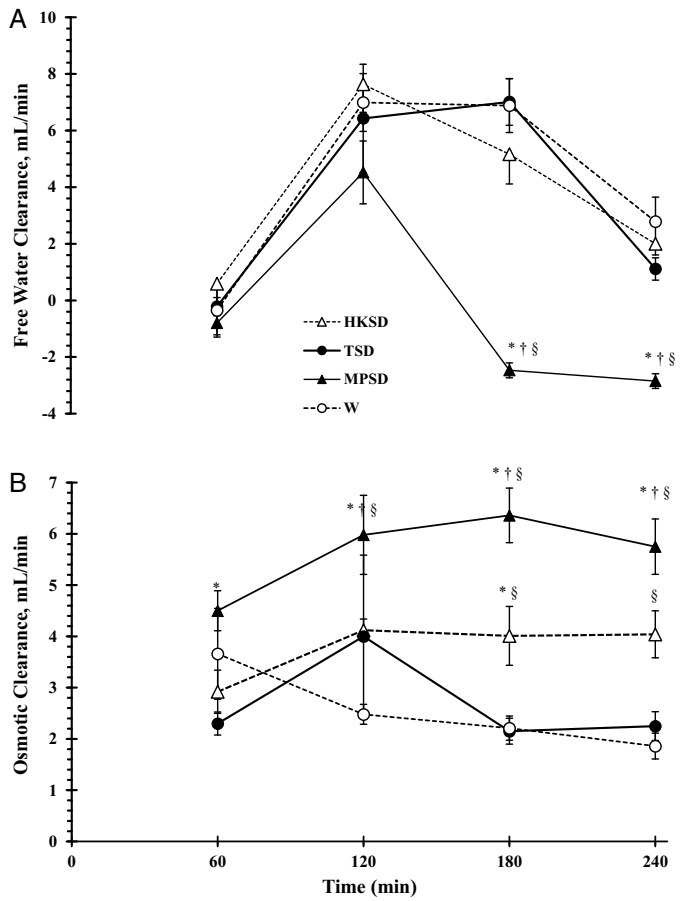

Figure 4 — Free water clearance (A) and osmotic clearance (B) during the 240 min of rehydration. Values represent mean ± *SD*. *Statistically significant difference from TSD. †Statistically significant difference from HKSD. §Statistically significant difference from W. TSD = traditional sports drink; HKSD = high-potassium sports drink; W = water.

($p < .05$). Similarly, cumulative Cl^- excretion (Table 3) showed a main effect of beverage, $F(3, 165) = 118.1$, $p < .001$; time, $F(3, 165) = 207$, $p < .001$; and Beverage × Time interaction, $F(9, 165) = 17$, $p < .001$. MPSD resulted in the highest cumulative Cl^- excretion, followed by HKSD ($p < .05$).

Urinary K^+ concentration was also significantly affected by beverage, $F(3, 259) = 9.3$, $p < .001$; time, $F(5, 259) = 20.6$, $p < .001$; and Beverage × Time interaction, $F(15, 259) = 4.9$, $p < .001$. W peaked at 60 min, exceeding MPSD, whereas TSD was lowest at 120 min compared with HKSD ($p < .05$). At 180 and 240 min, MPSD and HKSD showed elevated K^+ levels versus TSD and W, with MPSD peaking at 240 min above all others (Table 3; $p < .05$). A main effect of beverage, $F(3, 165) = 32.1$, $p < .001$; time, $F(3, 165) = 7.3$, $p < .001$; and a Beverage × Time interaction, $F(9, 165) = 9.2$, $p < .001$, was observed for K^+ excretion (Table 3). HKSD showed the highest K^+ excretion, higher than other beverages ($p < .05$). MPSD resulted in greater potassium excretion than TSD and W ($p < .05$). Similarly, there were main effects of beverage, $F(3, 165) = 66.6$, $p < .001$, and time, $F(3, 165) = 118.5$, $P < .001$, for cumulative K^+ excretion (Table 3), with a Beverage × Time interaction, $F(9, 165) = 12.6$, $p < .001$.

Plasma Osmolality and Volume

A main effect of beverage, $F(3, 351) = 56.1$, $p < .001$; time, $F(7, 351) = 7.7$, $p < .001$; and Beverage × Time interaction, $F(21, 351) = 4.1$, $p < .001$, on plasma osmolality was observed. MPSD elicited a sustained elevation in plasma osmolality throughout rehydration, exceeding HKSD and W at 60 min and remaining higher than all beverages at later time points (Figure 5A; $p < .05$). W was also lower than HKSD and TSD at 60, 90, and 120 min ($p < .05$).

Plasma volume was affected by beverage, $F(3, 301) = 7.5$, $p < .001$, and time, $F(6, 301) = 7.8$, $p < .001$. As shown in Figure 5C, TSD exhibited a consistently greater increase in plasma volume than MPSD and HKSD across the rehydration period ($p < .05$).

Plasma Electrolytes

A main effect of beverage, $F(3, 344) = 15.9$, $p < .001$; time, $F(7, 344) = 8.8$, $p < .001$; and Beverage × Time interaction, $F(21, 344) = 2.2$, $p = .003$, on plasma sodium concentration was observed. In the W trial, plasma sodium declined immediately postdehydration, reaching its lowest point at 120 min. Values were significantly lower than MPSD and HKSD at 60 min; MPSD, HKSD, and TSD at 90 and 120 min; and HKSD and TSD at 180 min. MPSD was also lower than HKSD at 120 and 180 min (Figure 5B; $p < .05$).

Beverage, $F(3, 343) = 60.4$, $p < .001$; time, $F(7, 343) = 13.2$, $p < 0.001$; and Beverage × Time interaction, $F(21, 343) = 5.9$, $p < .001$, significantly influenced plasma potassium concentration.

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Table 3 Urine Variables

Time, min	EUH	DEH	60	120	180	240
Urine sodium, mmol/L						
HKSD	32 ± 13	36 ± 26	55 ± 26	21 ± 6	31 ± 14	37 ± 14
TSD	49 ± 44	31 ± 23	54 ± 29	12 ± 3	14 ± 7	41 ± 33
MPSD	39 ± 35	35 ± 24	59 ± 25	29 ± 7	72 ± 16*,†,§	79 ± 21*,†,§
W	52 ± 52	51 ± 58	51 ± 29	12 ± 4	13 ± 6	28 ± 16
Sodium excretion, mmol						
HKSD			9 ± 4	14 ± 4	16 ± 8*,§	13 ± 8*,§
TSD			6 ± 3	8 ± 2	7 ± 3	6 ± 3
MPSD			11 ± 5	18 ± 7*,§	18 ± 7*,§	15 ± 6*,§
W			9 ± 8	7 ± 3	6 ± 3	5 ± 3
Cumulative sodium excretion, mmol						
HKSD			9 ± 4	24 ± 7	40 ± 13*,§	53 ± 19*,§
TSD			6 ± 3	14 ± 4	20 ± 6	27 ± 8
MPSD			11 ± 5	29 ± 12*,§	46 ± 13*,§	62 ± 14*,§
W			9 ± 8	16 ± 9	13 ± 22	26 ± 13
Urine potassium, mmol/L						
HKSD	11 ± 10	30 ± 20	42 ± 28	19 ± 9	36 ± 14*,§	47 ± 16*,§
TSD	19 ± 16	27 ± 15	33 ± 18	4 ± 2†	8 ± 4	27 ± 13
MPSD	13 ± 7	29 ± 23	35 ± 25§	16 ± 4	49 ± 12*,§	62 ± 20*,†,§
W	15 ± 10	43 ± 34	50 ± 35	11 ± 5	12 ± 12	21 ± 16
Potassium excretion, mmol						
HKSD			6 ± 3	12 ± 5*,§	18 ± 5*,§	15 ± 4*,§
TSD			4 ± 1	3 ± 1	4 ± 1	5 ± 3
MPSD			6 ± 3	10 ± 2*	12 ± 4*,†,§	11 ± 4*,§
W			9 ± 11	6 ± 2	4 ± 2	3 ± 1
Cumulative potassium excretion, mmol						
HKSD			6 ± 3	18 ± 7*	37 ± 10*,§	52 ± 12*,§
TSD			4 ± 1	6 ± 2	10 ± 3	15 ± 3
MPSD			6 ± 3	16 ± 5	28 ± 7*	39 ± 10*,†,§
W			9 ± 11	15 ± 12	13 ± 19	22 ± 14
Urine chloride, mmol/L						
HKSD	36 ± 15	55 ± 43	81 ± 39	30 ± 8	44 ± 14*	51 ± 13
TSD	50 ± 48	47 ± 30	75 ± 39	14 ± 4	15 ± 7	43 ± 27
MPSD	43 ± 38	55 ± 41	78 ± 36	38 ± 9	106 ± 21*,†,§	120 ± 21*,†,§
W	50 ± 48	78 ± 74	75 ± 35	17 ± 4	19 ± 14	36 ± 25
Chloride excretion, mmol						
HKSD			13 ± 4	20 ± 4*,§	23 ± 7*,§	18 ± 9*,§
TSD			8 ± 4	9 ± 2	8 ± 3	7 ± 4
MPSD			15 ± 6	24 ± 8*,§	25 ± 6*,§	22 ± 8*,§
W			13 ± 11	9 ± 3	7 ± 3	5 ± 3
Cumulative chloride excretion, mmol						
HKSD			13 ± 4	33 ± 6*	56 ± 11*,§	74 ± 16*,§
TSD			8 ± 4	17 ± 5	25 ± 7	32 ± 9
MPSD			15 ± 6	38 ± 13*,§	63 ± 15*,§	86 ± 15*,§
W			13 ± 11	22 ± 12	30 ± 14	35 ± 15
Urine osmolality, mmol/kg H_2O						
HKSD	223 ± 118	263 ± 167	337 ± 191	105 ± 36	152 ± 42	201 ± 33
TSD	323 ± 229	234 ± 97	335 ± 182	122 ± 199	79 ± 35	231 ± 113
MPSD	253 ± 190	268 ± 153	405 ± 146	204 ± 37§	482 ± 60*,†,§	563 ± 84*,†,§
W	320 ± 251	364 ± 245	395 ± 208	79 ± 32	108 ± 112	236 ± 202

Note. Values are represented as mean ± *SD*. EUH = euhydration baseline; DEH = dehydration baseline; HKSD = high-potassium sports drink; TSD = traditional sports drink; MPSD = milk permeate sports drink; W = water.
†Statistically significant difference compared with HKSD. *Statistically significant difference compared with TSD. §Statistically significant difference compared with W.

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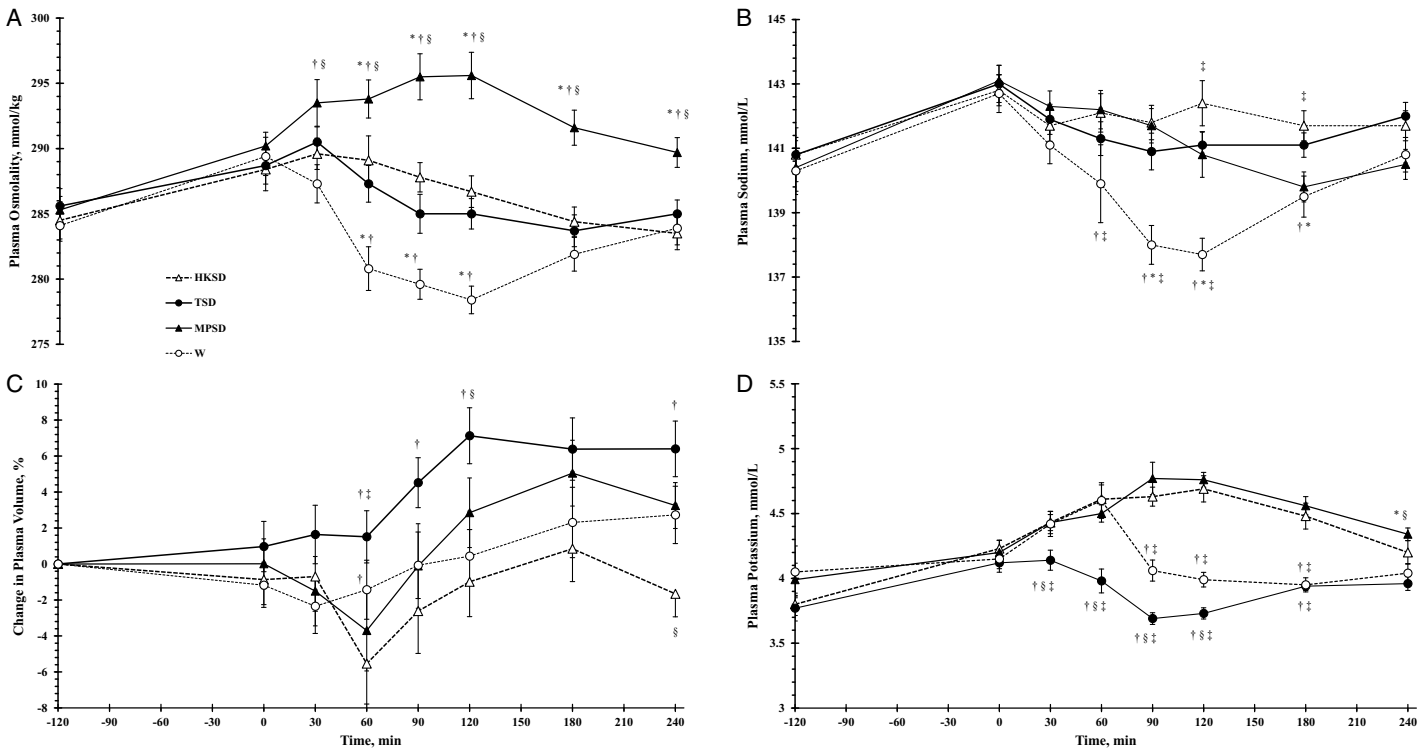

Figure 5 — Plasma osmolality (A), plasma sodium (B), percent change in plasma volume (C), and plasma potassium (D) throughout the experiment. Values represent mean ± *SD*. *Statistically significant difference from TSD. †Statistically significant difference from HKSD. §Statistically significant difference from W. ‡Statistically significant difference from MPSD. TSD = traditional sports drink; HKSD = high-potassium sports drink; W = water; MPSD = milk permeate sports drink.

At 30 min, TSD was lower than all others, a difference that persisted through 120 min and remained significant versus HKSD and MPSD at 180 min (Figure 5D; $p < .05$). W declined sharply between 60 and 90 min, resulting in lower values than MPSD and HKSD at 90, 120, and 180 min. At 240 min, MPSD was higher than W and TSD ($p < .05$).

Plasma chloride concentration was significantly influenced by beverage, $F(3, 336) = 24.5$, $p < .001$, and time, $F(7, 336) = 7.1$, $P < .001$. Overall, W resulted in lower plasma chloride levels than the other beverages (Table 4; $p < .05$).

Blood Lactate, Glucose, and Total Protein

Differences in blood lactate across beverages and time points were significant: beverage, $F(3, 342) = 93.9$, $p < .001$; time, $F(7, 342) = 69.3$, $p < .001$; and Beverage × Time interaction, $F(21, 342) = 11.3$, $p < .001$. W resulted in lower values than MPSD, TSD, and HKSD at 30, 60, 90, 120, and 180 min (Table 4; $p < .05$).

Differences in blood glucose across beverages and time points were significant: beverage, $F(3, 342) = 93.9$, $p < .001$; time, $F(7, 342) = 69.3$, $p < .001$; and Beverage × Time interaction, $F(21, 342) = 11.3$, $p < .001$. W resulted in lower values than MPSD, TSD, and HKSD at 30, 60, 90, 120, and 180 min (Table 4; $p < .05$), and MPSD exhibited lower glucose levels compared with TSD and HKSD at 30 min and lower than TSD at 60 and 90 min ($p < .05$). No significant difference was noticed in total protein (Table 4).

Perceptual Ratings

A significant effect of beverage, $F(3, 351) = 3.5$, $p = .016$, and time, $F(7, 351) = 62.1$, $p < .001$, on fullness was observed. Overall, MPSD resulted in greater fullness ratings compared with W and TSD (Table 5; $p < .05$). Taste perception was significantly affected by beverage, $F(3, 128) = 5.5$, $p = .0015$, and time, $F(2, 128) = 14.9$, $p < .001$. The taste score was consistently higher in the TSD trial compared with MPSD (Table 5; $p < .05$). No significant difference was noticed in the thirst perception (Table 5).

Discussion

This study demonstrated that following exercise-induced dehydration (−2.1% ± 0.4% body weight), MPSD showed a superior rehydration effectiveness compared with TSD, HKSD, and water when consumed at a volume equal to 150% of the lost body weight. NFB was significantly greater in MPSD compared with the other beverages at 180 and 240 min of rehydration ($p < .05$; Figure 1). These findings are consistent with previous research showing that milk-based beverages enhance fluid retention more than TSD (Desbrow et al., 2014; Shirreffs, Watson, et al., 2007; Volterman et al., 2014) and water (Aragón-Vargas et al., 2024) following exercise in the heat, likely due to reduced diuresis.

Sodium content is directly related to beverage rehydration effectiveness (Merson et al., 2008); however, sodium concentration of MPSD (20 mmol/L) is likely insufficient to restore fluid balance. Shirreffs et al. (2007) reported that adding 20 mmol/L sodium to milk (raising it to 58 mmol/L) did not result in any additional positive impact on fluid balance (Shirreffs, Watson, et al., 2007). Similarly, the addition of other electrolytes, such as potassium, has not consistently shown an apparent positive effect on fluid retention (Maughan et al., 1994; Pérez-Idárraga & Aragón-Vargas, 2014). These findings suggest that the electrolyte composition alone of the

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Table 4 Blood Variables

Time, min	EUH	DEH	30	60	90	120	180	240
Total protein, g/L								
HKSD	66 ± 6	68 ± 6	67 ± 07	70 ± 8	69 ± 8	68 ± 7	67 ± 7	67 ± 7
TSD	67 ± 5	68 ± 6	67 ± 7	68 ± 7	67 ± 10	65 ± 7	64 ± 7	65 ± 6
MPSD	65 ± 6	68 ± 6	68 ± 6	71 ± 8	69 ± 7	68 ± 6	65 ± 7	67 ± 7
W	66 ± 8	68 ± 7	68 ± 6	69 ± 9	66 ± 8	66 ± 8	65 ± 8	66 ± 7
Lactate, mmol/L								
HKSD	1.6 ± 0.6	1.1 ± 0.4	2 ± 0.4	2.9 ± 0.6[‡]	2.9 ± 0.6	2.5 ± 0.5	1.6 ± 0.4	0.9 ± 0.2
TSD	1.5 ± 0.7	1.1 ± 0.4	2 ± 0.6	2.9 ± 0.7	3.1 ± 0.8	2.5 ± 0.5	1.1 ± 0.3	0.8 ± 0.2
MPSD	1.4 ± 0.3	1.1 ± 0.3	1.4 ± 0.3	1.9 ± 0.4	2.0 ± 0.4	1.6 ± 0.4	1.1 ± 0.3	0.8 ± 0.2
W	1.5 ± 0.5	1.2 ± 0.3	1.1 ± 0.3[*,†,‡]	1.1 ± 0.3[*,†,‡]	0.9 ± 0.2[*,†,‡]	0.8 ± 0.2[*,†,‡]	0.8 ± 0.2[*,†,‡]	0.8 ± 0.2
Glucose, mmol/L								
HKSD	6.5 ± 1.1	5.9 ± 1	7.8 ± 0.9	6.3 ± 1.3	5.6 ± 0.9	5.0 ± 0.5	4.9 ± 0.6	4.6 ± 0.3
TSD	6.5 ± 0.9	5.7 ± 1	8.0 ± 1.0	6.8 ± 1.5	6.0 ± 1.3	5.1 ± 0.7	4.1 ± 0.8	4.5 ± 0.4
MPSD	6.6 ± 0.9	5.7 ± 1.1	7.0 ± 0.5[*,†]	5.8 ± 1.1[*]	5.3 ± 0.7[*]	5.5 ± 0.6	5.0 ± 0.4	4.9 ± 0.5
W	6.2 ± 0.6	5.8 ± 0.9	5.0 ± 0.6[*,†,‡]	5 ± 0.5[*,†,‡]	4.9 ± 0.4[*]	5.1 ± 0.4	4.9 ± 0.2	4.8 ± 0.3
Chloride, mmol/L								
HKSD	102 ± 3	106 ± 3	104 ± 3	104 ± 3	104 ± 3	105 ± 3	103 ± 3	104 ± 1
TSD	103 ± 2	106 ± 2	104 ± 2	103 ± 2	104 ± 2	104 ± 2	104 ± 2	104 ± 2
MPSD	102 ± 2	106 ± 2	104 ± 2	105 ± 2	105 ± 2	105 ± 2	104 ± 2	103 ± 2
W	102 ± 2	105 ± 2	102 ± 1[*,†,‡]	100 ± 2[*,†,‡]	100 ± 3[*,†,‡]	101 ± 2[*,†,‡]	102 ± 2[*,‡]	102 ± 2

Note. Values are represented as mean ± *SD*. EUH = euhydration baseline; DEH = dehydration baseline; HKSD = high-potassium sports drink; TSD = traditional sports drink; MPSD = milk permeate sports drink; W = water.
[†]Statistically significant difference compared with HKSD. [*]Statistically significant difference compared with TSD. [‡]Statistically significant difference compared with MPSD.

MPSD may not be the primary factor contributing to its superior hydration benefit.

Milk is recognized as a superior rehydration beverage due to its energy, fat, and protein content, which slows gastric emptying (Aragón-Vargas et al., 2024; Desbrow et al., 2014; James et al., 2011; Maughan et al., 2016). However, the MPSD lacked fat and protein and had lower energy and carbohydrate content than the sports drinks (Table 2). The superior rehydration observed with MPSD may be attributed to an increase in plasma osmolality (Figure 5A). Even a small elevation in plasma osmolality (2%–3%) can stimulate the secretion of arginine vasopressin, which suppresses diuresis and promotes renal water reabsorption (Stachenfeld, 2008). In the present study, plasma osmolality in the MPSD trial was significantly higher than in the TSD trial from 60 min onward; however, a corresponding reduction in urine output was not observed until after 120 min. The reason for this delayed urinary response remains unclear. Further research should clarify this lag, perhaps by measuring plasma copeptin, a stable surrogate marker of arginine vasopressin, aldosterone, and insulin.

The elevated plasma osmolality observed in MPSK may be attributed to beverage osmolality. Hypertonic beverages have been shown to delay gastric emptying (Vist & Maughan, 1995) and slow fluid retention in the proximal small intestine (Shi & Passe, 2010), thereby delaying water absorption into the circulation and preventing a rapid decline in plasma osmolality. Despite the slower fluid uptake, the early absorption of osmotically active components from MPSK may also contribute to an increase in plasma osmolality. This is supported by the greater osmotic excretion (Figure 3) and clearance (Figure 4) in MPSD early in rehydration. Having said that, electrolyte excretion results suggest an impaired electrolyte balance following MPSD (Table 3). Sodium excretion was higher in MPSD than in TSD despite comparable sodium content, and potassium excretion was similar to HKSD even though MPSD contained less potassium. These findings suggest that hydration benefits may have occurred even though there was greater electrolyte excretion.

The relatively higher osmolality of MPSD may be due to its extra minerals, like chloride, magnesium, iodine, and selenium, as well as erythritol (Table 2). Erythritol is a very low-calorie, nonnutritive sweetener that is mostly absorbed (>90%) in the small intestine via passive diffusion, not metabolized by human body, and excreted unchanged in the urine (Munro et al., 1998). Erythritol has a sweetness index of ~70% relative to sucrose and is less sweet than other nonnutritive sweeteners, allowing its use at higher concentrations in beverages. Erythritol administration induces gut hormones release and delays gastric emptying in a dose-dependent manner (Wölnerhanssen et al., 2016, 2021). Moreover, plasma erythritol peaks within 1–2 hr postingestion (Bornet et al., 1996); thus, its presence as a nonmetabolized osmotic solute may contribute to: (a) elevated plasma osmolality and (b) slower overall rate of fluid absorption. Although ~10% of ingested erythritol is not absorbed in the intestine (Munro et al., 1998) and could, in theory, promote water secretion into the lower gastrointestinal tract, particularly consumed in a hypertonic solution, no subjects in the present study reported defecation or experienced diarrhea. This is likely explained by the relatively low dosage administered (0.49 g/kg), which is below the level (~0.78 g/kg) associated with mild nausea and borborygmi (Storey et al., 2007).

Rapid expansion of extracellular water within the vascular compartment reflects a rapid water movement into the circulation (Evans et al., 2017; Maughan & Leiper, 1995). A transient reduction in plasma volume was observed in the MPSD trial compared

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Table 5 Perceptual Drink's Scores

Time, min	EUH	DEH	30	60	90	120	180	240
Thirst, mm								
HKSD	37 ± 30	96 ± 18.3	34 ± 15	15 ± 19	26 ± 19	36 ± 24	57 ± 22	75 ± 26
TSD	29 ± 15	104 ± 12	33 ± 19	15 ± 11	33 ± 21	43 ± 24	63 ± 23	69 ± 32
MPSD	31 ± 21	98 ± 14	40 ± 17	19 ± 22	28 ± 23	42 ± 30	67 ± 32	76 ± 30
W	29 ± 18	89 ± 23	37 ± 21	20 ± 22	29 ± 24	44 ± 27	54 ± 26	63 ± 32
Fullness, mm								
HKSD	71 ± 14	39 ± 19	75 ± 16	105 ± 22	83 ± 23	68 ± 29	44 ± 27	30 ± 27
TSD	72 ± 16	36 ± 19	86 ± 20	100 ± 18	82 ± 18	60 ± 23	43 ± 21	26 ± 23
MPSD	77 ± 15	33 ± 21	75 ± 19	108 ± 12	93 ± 24 §	81 ± 30*,§	52 ± 25	42 ± 32
W	76 ± 20	41 ± 23	73 ± 20	93 ± 23	72 ± 28	62 ± 27	40 ± 20	25 ± 15
Taste, mm								
HKSD	–	77 ± 23	66 ± 28	44 ± 37	–	–	–	–
TSD	–	100 ± 15	82 ± 19	63 ± 23	–	–	–	–
MPSD	–	70 ± 38*	56 ± 34*	40 ± 34	–	–	–	–
W	–	81 ± 24	63 ± 25	54 ± 35	–	–	–	–

Note. Values are represented as mean ± *SD*. EUH = euhydration baseline, DEH = dehydration baseline, HKSD = high-potassium sports drink, TSD = traditional sports drink, MPSD = milk permeate sports drink, and W = water.
*Statistically significant difference compared with TSD. §Statistically significant difference compared with W.

with TSD during the first hour of rehydration, which was subsequently restored by Minute 120 (Figure 5C). This paralleled blood glucose differences as MPSD resulted in lower levels than TSD at 60, 90, and 120 min (Table 4). Kamijo et al. (2012) demonstrated that adding carbohydrates to an electrolyte solution enhanced renal sodium reabsorption via insulin-mediated mechanisms, resulting in a greater plasma volume during the first 120 min compared with water, despite similar net fluid and sodium gains (Kamijo et al., 2012). We speculate that carbohydrate concentration and type (4% glucose/galactose in MPSD vs. 6% glucose/sucrose in TSD) may explain the differences in extracellular fluid retention. Galactose has a faster gastric emptying rate than glucose (Elias et al., 1968), shares intestinal transporters (Koepsell, 2020), and may enter into circulation more rapidly yet elicits a minimal glycemic and insulinogenic response (Ganda et al., 1979). In contrast, the higher glucose content in TSD likely induced a stronger insulin response, thereby enhancing sodium reabsorption and blood volume expansion relative to MPSD. Supporting this, urine sodium excretion tended to be higher in MPSD and was significantly elevated compared with TSD at 140 and 180 min. Despite these changes, plasma sodium concentrations remained similar between MPSD and TSD, indicating that the early reduction in plasma volume following MPSD ingestion occurred independently of changes in plasma sodium. One possible explanation is that the hypertonic nature of MPSD temporarily drew water from the vascular space into the intestinal lumen, reducing net water absorption and plasma volume during the early phase of rehydration.

It is essential to acknowledge the limitations of the present study. Fluid retention was considered an indicator of better rehydration; however, the proportion of retained fluid that remained in the stomach during rehydration was not assessed with the current methodology. Future research incorporating deuterium-labeled water could help determine the absorption speed of ingested fluid. In addition, measuring blood insulin levels would provide further insight into the role of carbohydrate intake in sodium reabsorption.

This is the first study to compare the rehydration efficacy of a milk-based sports drink with two TSDs following exercise in the heat. The novel milk permeate-based formulation, with an electrolyte profile similar to that of milk, facilitated a more sustained rehydration process. TSDs promote rapid plasma volume restoration but may not fully compensate for total water loss. The high-potassium beverage improved plasma electrolytes but failed to restore NFB or plasma volume due to higher urine excretion. Participants reported greater stomach fullness with MPSD than TSD at 120 min. This, combined with its lower carbohydrate and energy content and more favorable fluid balance, suggests that MPSD may be a suitable option for rehydration, particularly in situations where reduced food intake is desirable.

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Minerals, and Rockley Photonics. He has also provided scientific consultation to the American Beverage Association, Unilever, Pepsico, The Coca-Cola Company, and Kraft Heinz. He is also the owner of Dr Hydration LLC. **Funding Sources**: The study was funded by a grant from the National Dairy Council. **Protocol**: The study was registered before the start of recruitment at ClinicalTrials.gov under the number NCT06415409.

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Nontechnical Summary

When people exercise, especially in hot conditions, their bodies lose water through sweat. This is the body's normal way of cooling itself, but it can lead to dehydration if the lost fluids are not replaced. Most people do not drink enough during or after exercise, so they often remain dehydrated for several hours. That is why it is important to find drinks that can help the body rehydrate effectively after exercise.

Rehydration is not just about drinking water. The body replaces fluids better when the drink includes the right mix of ingredients. These include sodium (a type of salt), carbohydrates (a form of energy), and other minerals. Most sports drinks are designed to include these ingredients, but many of them do not contain enough sodium to fully replace the sodium lost through sweat. Milk has been shown to be more effective than traditional sports drinks (TSDs) in some cases, but some people do not tolerate milk well, and it is higher in calories.

In this study, we tested a newer type of drink made from milk permeate. Milk permeate is a filtered component of milk that contains important minerals, such as sodium and potassium, but lacks protein, fat, and lactose. This milk permeate-based drink was created to provide the hydration benefits of milk without the heaviness or extra calories.

To test the effectiveness of this drink, participants exercised in the heat until they lost approximately 2% of their body weight through sweat. This amount of dehydration is enough to affect performance and recovery. After exercising, they drank one of four different beverages:

- the milk permeate-based sports drink
- a TSD
- a high-potassium sports drink
- plain water

Each participant drank enough fluid to replace 150% of the weight they lost. This amount helps ensure full rehydration as when you start drinking, your body also loses fluids via urination.

The researchers measured the amount of fluid retained in the body over the next 4 hr. This was called net fluid balance, and it indicated how much fluid was retained within the body. The researchers also examined blood levels, urine output, and participants' feelings of fullness.

What the Study Found

The milk permeate-based drink helped people hold onto more fluid than the TSD, the high-potassium drink, or plain water, especially after 3 and 4 hr of rehydration.

This result was surprising because the milk permeate-based drink did not have more sodium than the TSD. Sodium is known to help with fluid retention, so something else must have made this drink more effective.

One likely reason is that the milk permeate-based drink had a higher concentration of dissolved particles. This slows down the rate at which the stomach empties and delays fluid absorption, which helps keep fluids in the body longer and reduces the amount of urine the body produces.

Another possible reason is the use of erythritol, an ingredient that is not digested and is excreted from the body through urine. This sweetener might help hold water in the body temporarily and prevent it from being lost too quickly.

The study also found that people had higher levels of certain minerals in their blood after drinking the milk permeate-based drink. Their bodies also produced less urine in the first part of the recovery period.

Although fluid absorption was slower, the milk permeate-based drink helped people maintain better hydration overall. This may be because it helped maintain the concentration of substances in the blood, which signals the body to hold onto more water.

Why This Matters

This is the first study to investigate the comparison between a milk permeate-based sports drink and TSD in terms of their effects after exercise in the heat. The results suggest that the milk permeate drink is a suitable option for rehydration, particularly when the goal is to maintain hydration over several hours.

For athletes or those who exercise regularly, this drink may be a good choice. It provides important minerals and helps the body hold onto water without the extra calories or stomach discomfort that may come with milk or high-sugar drinks.

Although further research is needed to fully understand how the drink works within the body, this study suggests that the milk permeate-based sports drink can be a useful tool for postexercise recovery. It helps fill a gap between water and TSDs, especially in situations where staying hydrated over time is important.

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